

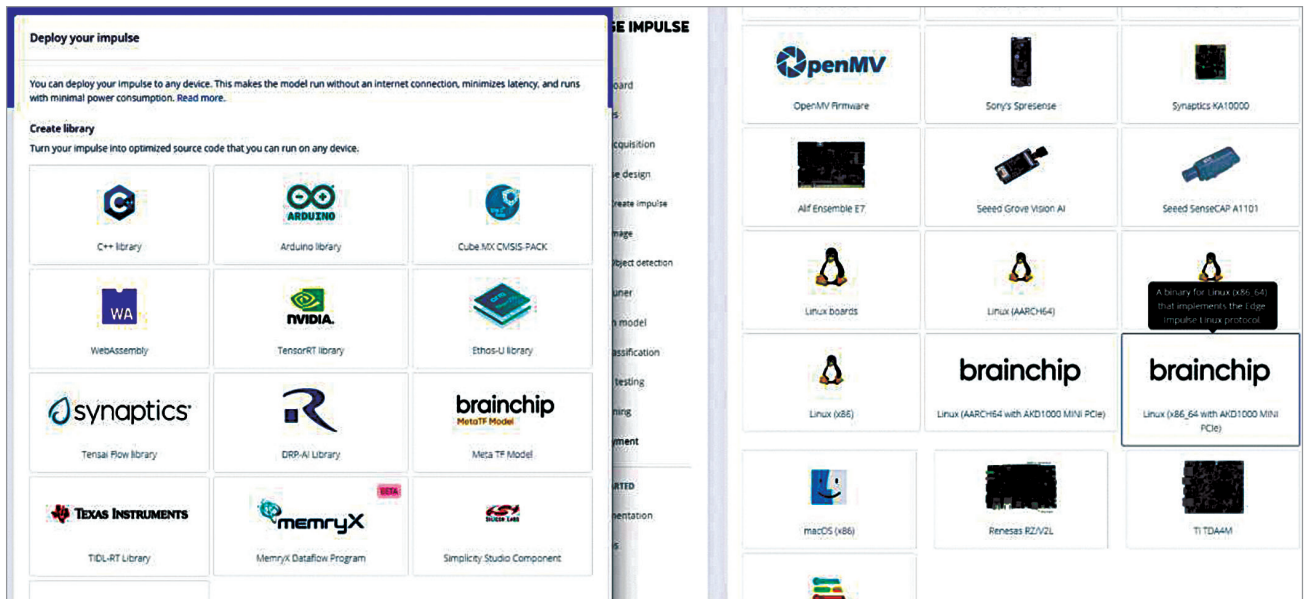


Fig. 5: Exporting model code for deployment

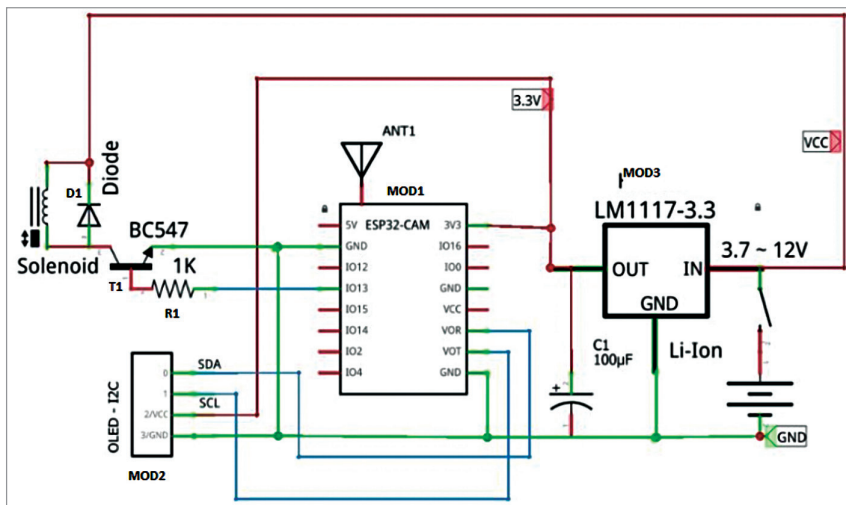


Fig. 6: Circuit diagram

tion. However, YOLO and FOMO are easier and work in the most acceptable levels for object classification models. Select the model and then press it for training. This process takes some time, so relax. Fig. 3 shows the training ML model.

6. At the end of training, check for the 'F1 score,' which should be 85% or above. To improve the score, you may have to change the model or remove some of the outlier images that may have been deteriorating the overall score.

7. To test the model, 20% of the data that was set aside may now be used. Just click on each image individually or all of them together. Accuracy should be well within the accepted level (81% to 91%). However, 100% accuracy is not good for the model. In that case we have to insert some error intentionally! Fig. 4 shows the testing model.

Deployment of model

Now after testing the ML model, we can deploy it on many boards.

As we are using ESP32 cam, select the Arduino IDE. Fig. 5 shows the exporting of model code for deployment.

After selecting the model (Arduino), press 'Build' button at the bottom. The Arduino sketch, along with the necessary library, will be downloaded on your local computer. In Arduino IDE, use this zip file to install it as a new library (Sketch→Include library→Add .zip library...).

Once the library is installed, go to file→library→find the latest library→examples→esp32_camera→your-sketch-is-here

Uploading the sketch

The model under ESP32 is all set for ESP-EYE camera board. The low-cost ESP32 camera available in the open market is usually the 'ESP32 AI Thinker cam' or a little costlier 'ESP32 TTGO T plus camera' model. For both of these we have set the pin details and inserted those camera models in the sketch software. You just have to uncomment the right model and the sketch is all set to get installed.

Light is required during the